Otto-von-Guericke-University

Magdeburg

Faculty of Process and Systems Engineering

Legal Issues in Plant Operations and Process Safety – Safety Report SS 2024

"SAFETY STUDY FOR A PRODUCTION FACILITY UNDER THE SEVESO-III DIRECTIVE "

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EXECUTIVE SUMMARY

This study focuses on the safety of a chemical production facility about relation to the Seveso-III Directive (Directive 2012/18/EU), which imposes strict safety protocols for establishments that handle large quantities of hazardous substances. The facility stores a variety of dangerous chemicals, such as natural gas, gasoline, biogas, ethanol, acetylene, and hydrogen peroxide, all of which pose significant risks of fire, explosion, and toxic release. Our comprehensive analysis categorizes the facility as an upper-tier establishment under the Seveso-III Directive due to the quantities of these hazardous substances. To assess potential accidents, we utilized the ALOHA software to simulate scenarios involving gas leaks, fires, and explosions. This allowed us to identify critical areas that are at risk of domino effects and secondary incidents. The study emphasizes the importance of existing preventive measures, such as advanced fire suppression systems, continuous leak detection, and strict process controls. However, it also highlights the need for ongoing review and improvement of these measures to address evolving risks.

In addition, we evaluated the vulnerability of the surrounding areas, including residential zones, industrial facilities, and environmentally sensitive regions, to major accidents. This underscores the importance of robust evacuation plans and community awareness programs. The report recommends the enhancement of monitoring technologies, regular risk assessments, increased community engagement, continuous training for personnel, and investment in infrastructure upgrades to enhance safety and resilience. By adhering to the Seveso-III Directive and implementing these recommendations, the facility can significantly reduce the risk of major accidents. This not only ensures the safety of the facility itself but also protects the surrounding community and demonstrates a strong commitment to safeguarding human health and environmental integrity.

TABLE OF CONTENTS

LIST OF ABBREVIATIONS

EU	European Union
ERPG	Emergency Response Planning Guidelines
ALOHA	Areal Locations of Hazardous Atmospheres
PPE	Personal Protective Equipment

LIST OF FIGURES

LIST OF TABLES

1. INTRODUCTION

The chemical production industry is integral to modern society, providing essential products ranging from fuels to pharmaceuticals. However, this industry also involves the handling of hazardous substances, posing significant risks to human health and the environment. To mitigate these risks, the European Union has established the Seveso-III-Directive, formally known as Directive 2012/18/EU on the control of major-accident hazards involving dangerous substances. This directive aims to prevent major accidents and limit their consequences, ensuring a high level of protection throughout the European Union.

This safety study is prepared in compliance with the Seveso-III-Directive, specifically focusing on a chemical production facility that handles a variety of hazardous substances. The facility's inventory includes substantial quantities of natural gas, gasoline, biogas, ethanol, self-decomposing substances, and oxidizing liquids, all of which are subject to strict regulatory oversight due to their potential to cause significant harm in the event of an accident.

The purpose of this safety study is to evaluate the risks and necessary safety measures for a chemical production facility in compliance with the Seveso-III-Directive. This directive aims to prevent major accidents involving dangerous substances and to limit the consequences of such accidents on human health and the environment.

The following points are explained further:

1. **Classification of the Facility**: Determining whether the facility qualifies as a lower-tier or upper-tier establishment under the Seveso-III-Directive based on the quantities of hazardous substances present.
2. **Risk Assessment**: Identifying and analysing potential accidental risks, including fire, explosion, and toxic release scenarios using ALOHA Software.
3. **Prevention and Mitigation Measures:** Outlining the measures in place to prevent accidents and mitigate their consequences, including process controls, safety training, emergency response plans, and coordination with local emergency services.
4. **Vulnerability Analysis**: Evaluating the vulnerability of the surrounding area to major accidents, including the potential impact on residential areas, industrial facilities, and sensitive environments.

2. DESCRIPTION OF THE ENVIRONMENT OF THE ESTABLISHMENT, PLANT, AND PROCESS

2.1 Description of Environment of Establishment Area

The chemical production facility is strategically located in an industrial zone to optimize logistical operations and minimize the impact on residential areas. This section provides a detailed description of the establishment area, focusing on the following aspects:

- **Geographical Location**: The central part of the image shows a dense area with various industrial facilities which include large storage tanks used for storage of liquids or gases and other Manufacturing facilities. A significant river ("River Main") runs beside the industrial facility. There are few ships / barges visible on the river which indicates that it is a navigable waterway used for transporting goods. There is also a dedicated area for shipping containers and cargo handling through the waterway indicated as a port or logistics. There is also a bridge that crosses the river which shows a connection from the Industrial area to the residential or commercial area for transportation of goods as well. The other side of the industrial facility is the Residential/ commercial neighbourhood's with buildings and greenery. From this information it is clear that the site is well-connected by road and water networks, facilitating the transportation of raw materials and finished products

2.2 Description of Plant and Process Steps

The Chemical Production facility is designed with multiple blocks, each dedicated to specific operations.

Plant & Processing units: The production facility involves various blocks, such as raw material preparation, chemical reactions, packaging, Separation, Distillations, Purification, etc., Each stage is monitored carefully through automation and mechanical systems to ensure enhanced safety and efficiency. The facility also includes dedicated storage tanks for Natural gas, Gasoline, Biogas, Ethanol, Self-decomposing substances, and oxidizing liquids.

2.3 Overview of Hazards of Hazardous Materials

The facility contains a variety of hazardous materials, each one has its own set of potential to cause risks and hazards.

- ***Natural gas:***
  Danger: Poses a risk of fire and explosion
  Control: Proper Ventilation and leak detection systems are necessary to mitigate this risk.
- ***Gasoline:***
  Danger: Highly flammable and can also form explosive mixtures with air.
  Control: Currently storage tanks are equipped with floating roofs to minimise vapor loss & also potential ignition sources are strictly controlled.
- ***Biogas***:
  Danger: Primarily flammable and highly explosive.

Control: Ensure adequate ventilation in storage tanks.

- ***Other flammable liquids (Ethanol):***
  Danger: Highly flammable, risk of explosion in confined spaces.
  Control: Storage in properly designed tanks with appropriate venting.
- ***Self-decomposing substances (Acetylene):***
  Danger: Risk of self-heating and subsequent fire or explosion.
  Control: Continuous monitoring of storage conditions.
- ***Oxidising liquids (Hydrogen Peroxide):***
  Danger: Can intensify fires and increase the combustion rate of other materials.
  Control: Storage should be away from flammable and combustible materials.

3. DESCRIPTION OF THE INSTALLATION

Determining Tier Classification: The Seveso-III Directive categorizes establishments as either lower-tier or upper-tier based on the quantities of hazardous substances present. To determine the classification of the facility, we need to compare the substance inventories against the threshold quantities specified in Annex I of the Directive.

3.1 Inventory Analysis

1. ***Natural Gas: Amount: 15,000 m³ at 7 bar*** [1]

Conversion:

- Density of natural gas (approx.): 0.7 kg/m³
- Total mass: 26,423kg (using ideal gas law equation)
- Total mass in tonnes: 26,423 kg÷1,000=26.423 tonnes.

2. ***Gasoline: Amount: 25,000 m3*** [1]

Conversion:

- Density of gasoline (approx.): 0.79 kg/l
- Total mass: 25,000 m3 × 0.79 kg/L × 1,000 L/m3 = 19,750,000
- Total mass in tonnes: 19,750,000 kg ÷ 1,000=19,750 tonnes.

3. ***Biogas: Amount: 8,000 m³ at 1 bar*** [1]

Conversion:

- Density of biogas (approx.): 1.2 kg/m³
- Total mass: 8,000 m3 × 1.2 kg/m3 = 9,600 kg
- Total mass in tonnes: 9,600 kg ÷ 1,000 = 9.6 tonnes.

4. ***Other Flammable Liquids (Ethanol): Amount: 40,000 m³*** [1]

Conversion:

- Density of ethanol (approx.): 0.789 kg/L
- Total mass: 40,000 m3 × 0.789 kg/L × 1,000 L/m3 = 31,560,000 kg.
- Total mass in tonnes: 31,560,000 kg ÷ 1,000 = 31,560 tonnes

5. ***<u>Self-Decomposing Substances: Amount: 400 tons</u>*** [1]
   Total mass in tonnes: 400*0.907185 = 362.874 tonnes

6. ***<u>Oxidizing Liquids: Amount: 200 tons</u>*** [1]
   Total mass in tonnes: 200*0.907185 = 181.437 tonnes

3.2 Comparison with Thresholds

Table 1 Overview of Inventory and Tier Classification

Substance Name	**Inventory (Tonnes)**	**Lower-tier (Tonnes)**	**Upper-tier (tonnes)**	**Classification**
Natural Gas	26.243	50	200	Lower - tier
Gasoline	19,750	2,500	25,000	Upper- tier
Biogas	9.6	50	200	Below lower-tier
Other Flammable Liquids (Ethanol)	31,560	5000	50,000	Upper-tier
Self-Decomposing Substances	362.874	5	50	Upper-tier
Oxidizing Liquids	181.437	50	200	Upper-tier

3.3 Summation of upper-tier Establishment

Sum = QU1*q1 + QU2*q2 + QU3*q3 + QU4*q4 + QU5*q5 + QU6*q6…………..

Where qx = the quantity of dangerous substance [2]

Q UX = the relevant qualifying quantity for dangerous substance

= ((26.243/200) +(19750/25000) +(9.6/200) +(31560/50000) +(362.874/50) +(181.437/200))

= 9.76508 (which is >1).

3.4 Conclusion

The Summation value obtained by using above formula is 9.67508 which is greater than 1, So, the facility contains Hazardous Substances that exceed the upper-tier thresholds (Natural gas, Ethanol, Self-Decomposing Substances & Oxidising Liquids). So, the scenario is classified as Upper-tier Establishment under Seveso-III Directive.

4. IDENTIFICATION AND ACCIDENTAL RISKS ANALYSIS AND PREVENTION METHODS

4.1 Major Accident Scenarios

The possible major accident scenarios include

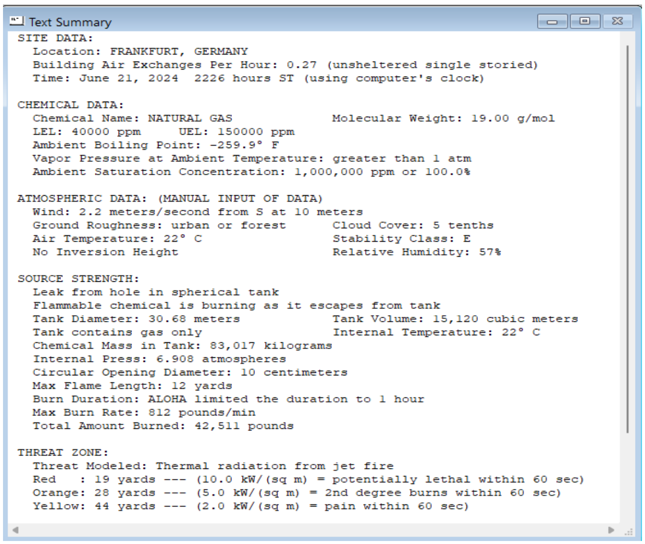
- Explosions or fires in storage tanks.
- Chemical spills or leaks during processing or transportation.
- Domino effects from accidents in neighbouring establishments.

This scenario causes severe damage & harm to the human health, environment and the facilities, particularly in nearby residential areas, industrial facilities and water bodies.

4.2 Risk analysis and pressure wave impact of the substances

4.2.1 Identified Risks and Hazards:

1. ***Natural Gas:*** [3] [4] [5] [1] [6] [7]
   Thermal Radiation: The Red zone which has an impacted area till 19 yards & Orange zone impact which has an impacted area till 28 yards and yellow zone which has an impacted area till 44 yards.
   Explosion risk: As Natural gas by nature has the potentiality to poses a risk of fire and explosion. Where its explosion limits lie between 4% - 15%.
   Health Hazards: Inhalation of larger concentrations of Natural gas has the ability to displace oxygen which in turn leads to asphyxiation.
   Environmental impact: Combustion of Natural gas release pollutants this can cause air pollution and also damage to the sites.



Text Summary

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SITE DATA:
   Location: FRANKFURT, GERMANY
   Building Air Exchanges Per Hour: 0.27 (unsheltered single storied)
   Time: June 21, 2024  2226 hours ST (using computer's clock)

CHEMICAL DATA:
   Chemical Name: NATURAL GAS              Molecular Weight: 19.00 g/mol
   LEL: 40000 ppm     UEL: 150000 ppm
   Ambient Boiling Point: -259.9° F
   Vapor Pressure at Ambient Temperature: greater than 1 atm
   Ambient Saturation Concentration: 1,000,000 ppm or 100.0%

ATMOSPHERIC DATA: (MANUAL INPUT OF DATA)
   Wind: 2.2 meters/second from S at 10 meters
   Ground Roughness: urban or forest       Cloud Cover: 5 tenths
   Air Temperature: 22° C                  Stability Class: E
   No Inversion Height                     Relative Humidity: 57%

SOURCE STRENGTH:
   Leak from hole in spherical tank
   Flammable chemical is burning as it escapes from tank
   Tank Diameter: 30.68 meters             Tank Volume: 15,120 cubic meters
   Tank contains gas only                  Internal Temperature: 22° C
   Chemical Mass in Tank: 83,017 kilograms
   Internal Press: 6.908 atmospheres
   Circular Opening Diameter: 10 centimeters
   Max Flame Length: 12 yards
   Burn Duration: ALOHA limited the duration to 1 hour
   Max Burn Rate: 812 pounds/min
   Total Amount Burned: 42,511 pounds

THREAT ZONE:
   Threat Modeled: Thermal radiation from jet fire
   Red   : 19 yards --- (10.0 kW/(sq m) = potentially lethal within 60 sec)
   Orange: 28 yards --- (5.0 kW/(sq m) = 2nd degree burns within 60 sec)
   Yellow: 44 yards --- (2.0 kW/(sq m) = pain within 60 sec)
```

Figure 1 Natural gas data file using ALOHA software

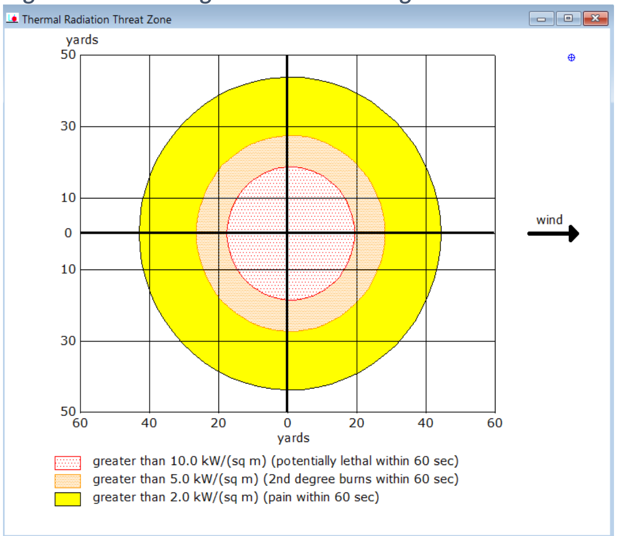


Figure 2 Natural gas_ALOHA_Threatzone_Case_1

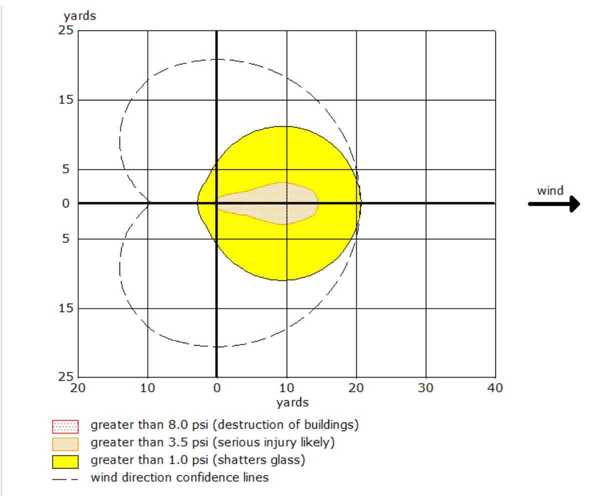
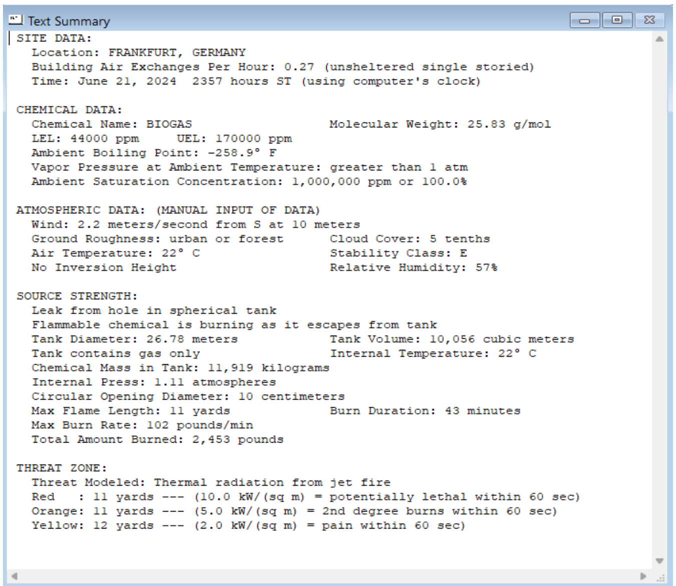


Figure 3 Naturalgas_ALOHA_Threatzone_Case_2

2. ***Biogas:*** [4] [5] [1] [7] [8] [6]
Thermal Radiation: The Red zone which has an impacted area till 11 yards & Orange zone impact which has an impacted area till 11 yards and yellow zone which has an impacted area till 12 yards. The orange and yellow zones could occur within the same time frame.
Explosion risk: The high internal pressure which has the potential for a more vigorous explosion in the ambient environment. The explosion limits lie between 4% - 17%.
Health Hazards: Inhalation of larger concentrations of Biogas has the ability to displace oxygen which in turn leads to asphyxiation.
Environmental impact: Combustion of Biogas release pollutants this can cause air pollution and also damage to the sites.



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Text Summary
SITE DATA:
  Location: FRANKFURT, GERMANY
  Building Air Exchanges Per Hour: 0.27 (unsheltered single storied)
  Time: June 21, 2024  2357 hours ST (using computer's clock)

CHEMICAL DATA:
  Chemical Name: BIOGAS                    Molecular Weight: 25.83 g/mol
  LEL: 44000 ppm     UEL: 170000 ppm
  Ambient Boiling Point: -258.9° F
  Vapor Pressure at Ambient Temperature: greater than 1 atm
  Ambient Saturation Concentration: 1,000,000 ppm or 100.0%

ATMOSPHERIC DATA: (MANUAL INPUT OF DATA)
  Wind: 2.2 meters/second from S at 10 meters
  Ground Roughness: urban or forest        Cloud Cover: 5 tenths
  Air Temperature: 22° C                   Stability Class: E
  No Inversion Height                      Relative Humidity: 57%

SOURCE STRENGTH:
  Leak from hole in spherical tank
  Flammable chemical is burning as it escapes from tank
  Tank Diameter: 26.78 meters              Tank Volume: 10,056 cubic meters
  Tank contains gas only                   Internal Temperature: 22° C
  Chemical Mass in Tank: 11,919 kilograms
  Internal Press: 1.11 atmospheres
  Circular Opening Diameter: 10 centimeters
  Max Flame Length: 11 yards               Burn Duration: 43 minutes
  Max Burn Rate: 102 pounds/min
  Total Amount Burned: 2,453 pounds

THREAT ZONE:
  Threat Modeled: Thermal radiation from jet fire
  Red   : 11 yards --- (10.0 kW/(sq m) = potentially lethal within 60 sec)
  Orange: 11 yards --- (5.0 kW/(sq m) = 2nd degree burns within 60 sec)
  Yellow: 12 yards --- (2.0 kW/(sq m) = pain within 60 sec)
```

Figure 4 Biogas _ALOHA Data file

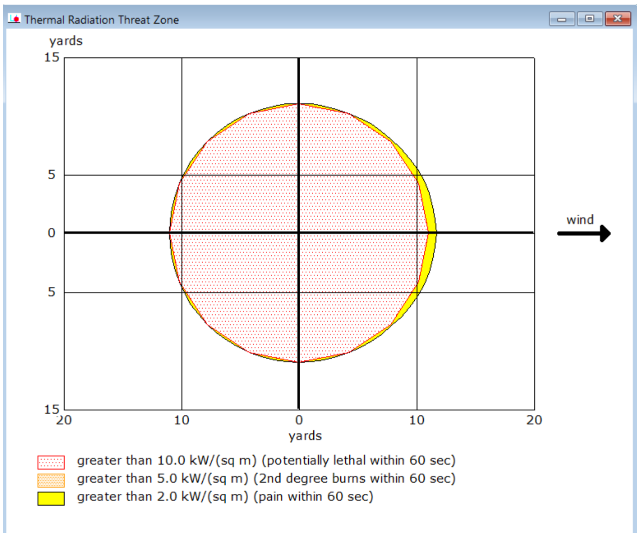


Figure 5 Biogas _ALOHA_ Threat zone_Case_1

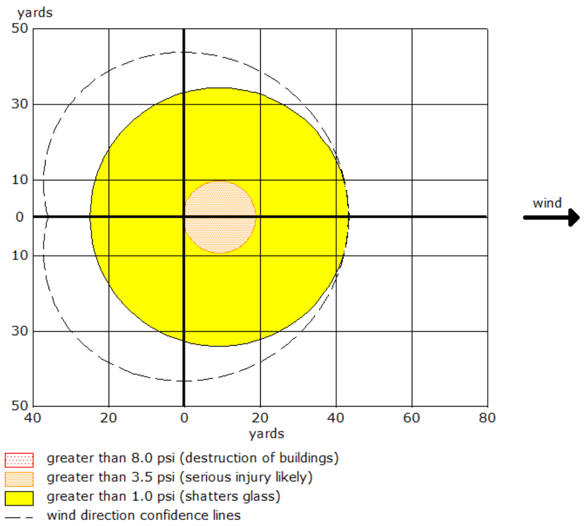
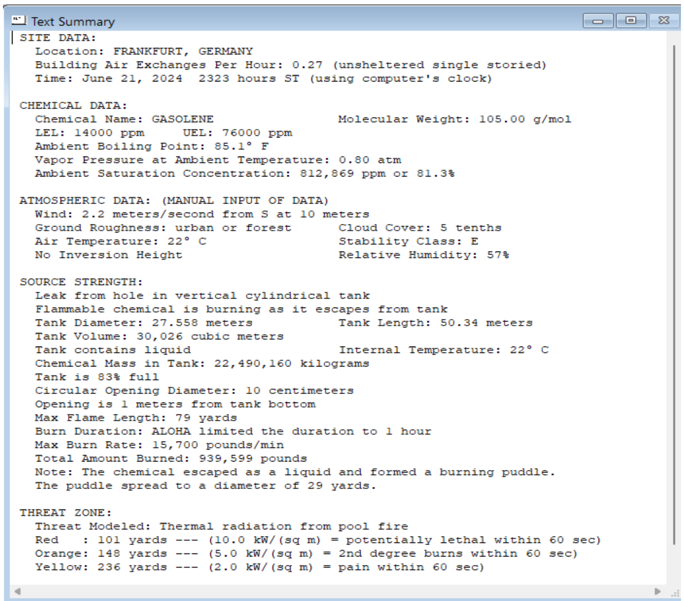


Figure 6 Biogas_ALOHA_Threatzone_Case_2

3. ***Gasoline:*** [3] [4] [5] [6] [7] [9] [1]
   Thermal Radiation: The Red zone which has an impacted area till 101 yards & Orange zone impact which has an impacted area till 148 yards and yellow zone which has an impacted area till 236 yards.
   Explosion risk: The high internal pressure which has the potential for a more vigorous explosion in the ambient environment. The explosion limits lie between 1.4% - 7.6%.
   Health Hazards: Inhalation of gasoline vapours can cause dizziness, nausea and in high concentrations may lead to asphyxiation.
   Environmental impact: Combustion of gasoline release pollutants such as carbon monoxide and volatile organic compounds can cause air pollution and also potential environmental damage.



Text Summary
SITE DATA:
Location: FRANKFURT, GERMANY
Building Air Exchanges Per Hour: 0.27 (unsheltered single storied)
Time: June 21, 2024 2323 hours ST (using computer's clock)

CHEMICAL DATA:
Chemical Name: GASOLENE Molecular Weight: 105.00 g/mol
LEL: 14000 ppm UEL: 76000 ppm
Ambient Boiling Point: 85.1° F
Vapor Pressure at Ambient Temperature: 0.80 atm
Ambient Saturation Concentration: 812,869 ppm or 81.3%

ATMOSPHERIC DATA: (MANUAL INPUT OF DATA)
Wind: 2.2 meters/second from S at 10 meters
Ground Roughness: urban or forest Cloud Cover: 5 tenths
Air Temperature: 22° C Stability Class: E
No Inversion Height Relative Humidity: 57%

SOURCE STRENGTH:
Leak from hole in vertical cylindrical tank
Flammable chemical is burning as it escapes from tank
Tank Diameter: 27.558 meters Tank Length: 50.34 meters
Tank Volume: 30,026 cubic meters
Tank contains liquid Internal Temperature: 22° C
Chemical Mass in Tank: 22,490,160 kilograms
Tank is 83% full
Circular Opening Diameter: 10 centimeters
Opening is 1 meters from tank bottom
Max Flame Length: 79 yards
Burn Duration: ALOHA limited the duration to 1 hour
Max Burn Rate: 15,700 pounds/min
Total Amount Burned: 939,599 pounds
Note: The chemical escaped as a liquid and formed a burning puddle.
The puddle spread to a diameter of 29 yards.

THREAT ZONE:
Threat Modeled: Thermal radiation from pool fire
Red : 101 yards --- (10.0 kW/(sq m) = potentially lethal within 60 sec)
Orange: 148 yards --- (5.0 kW/(sq m) = 2nd degree burns within 60 sec)
Yellow: 236 yards --- (2.0 kW/(sq m) = pain within 60 sec)

Figure 7 Gasoline _ALOHA _Datafile

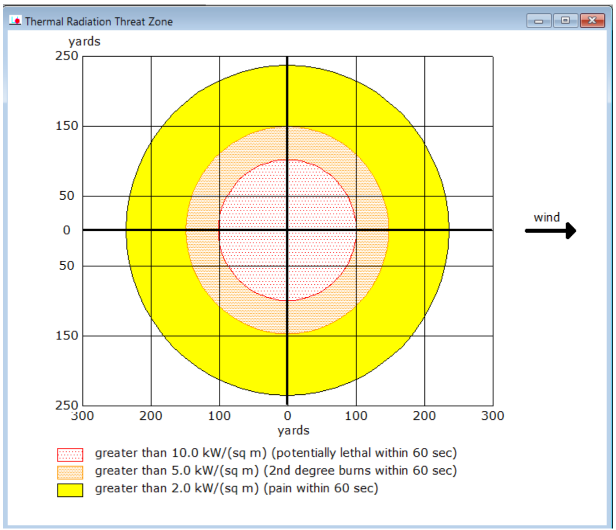


Figure 8 Gasoline _ ALOHA _ Threat Zone_Case_1

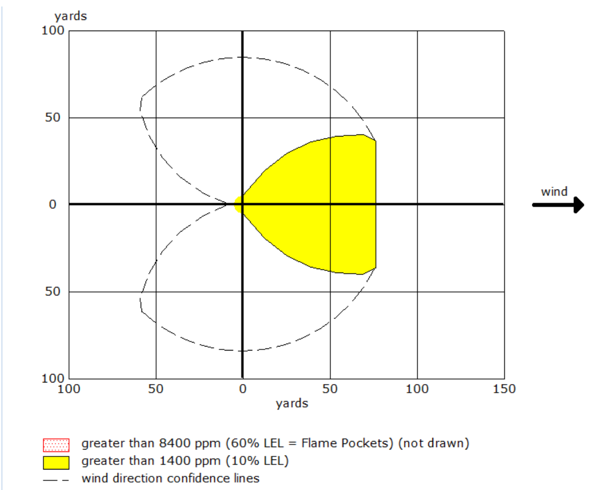


Figure 9 Gasoline_ALOHA_Threatzone_Case_2

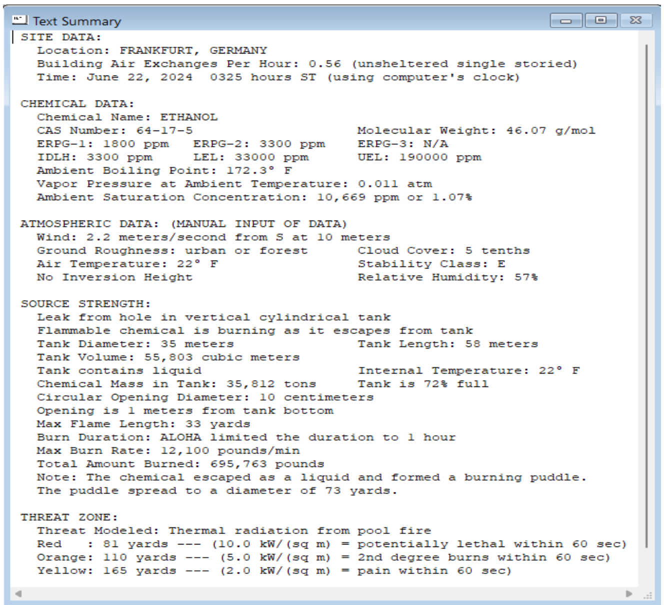
4. ***Ethanol:*** [10] [4] [1] [5] [7] [6]

Thermal Radiation: The Red zone which has an impacted area till 81 yards & Orange zone impact which has an impacted area till 110 yards and yellow zone which has an impacted area till 164 yards.

Explosion risk: Ethanol does not have specified explosive limits but its flammability makes it prone to fire and explosion under certain conditions.

Health Hazards: Inhalation of ethanol vapours can cause respiratory irritation and dizziness and also higher concentrations can lead to central nervous system depression.

Environmental impact: Combustion of ethanol release pollutants such as carbon monoxide and volatile organic compounds can cause air pollution and also potential environmental damage. Spilled ethanol can contaminate soil and water posing risks to aquatic ecosystems.



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Text Summary
SITE DATA:
   Location: FRANKFURT, GERMANY
   Building Air Exchanges Per Hour: 0.56 (unsheltered single storied)
   Time: June 22, 2024  0325 hours ST (using computer's clock)

CHEMICAL DATA:
   Chemical Name: ETHANOL
   CAS Number: 64-17-5                        Molecular Weight: 46.07 g/mol
   ERPG-1: 1800 ppm   ERPG-2: 3300 ppm    ERPG-3: N/A
   IDLH: 3300 ppm     LEL: 33000 ppm      UEL: 190000 ppm
   Ambient Boiling Point: 172.3° F
   Vapor Pressure at Ambient Temperature: 0.011 atm
   Ambient Saturation Concentration: 10,669 ppm or 1.07%

ATMOSPHERIC DATA: (MANUAL INPUT OF DATA)
   Wind: 2.2 meters/second from S at 10 meters
   Ground Roughness: urban or forest       Cloud Cover: 5 tenths
   Air Temperature: 22° F                  Stability Class: E
   No Inversion Height                     Relative Humidity: 57%

SOURCE STRENGTH:
   Leak from hole in vertical cylindrical tank
   Flammable chemical is burning as it escapes from tank
   Tank Diameter: 35 meters                Tank Length: 58 meters
   Tank Volume: 55,803 cubic meters
   Tank contains liquid                    Internal Temperature: 22° F
   Chemical Mass in Tank: 35,812 tons      Tank is 72% full
   Circular Opening Diameter: 10 centimeters
   Opening is 1 meters from tank bottom
   Max Flame Length: 33 yards
   Burn Duration: ALOHA limited the duration to 1 hour
   Max Burn Rate: 12,100 pounds/min
   Total Amount Burned: 695,763 pounds
   Note: The chemical escaped as a liquid and formed a burning puddle.
   The puddle spread to a diameter of 73 yards.

THREAT ZONE:
   Threat Modeled: Thermal radiation from pool fire
   Red   : 81 yards --- (10.0 kW/(sq m) = potentially lethal within 60 sec)
   Orange: 110 yards --- (5.0 kW/(sq m) = 2nd degree burns within 60 sec)
   Yellow: 165 yards --- (2.0 kW/(sq m) = pain within 60 sec)
```

Figure 10 Ethanol_ ALOHA_ Datafile

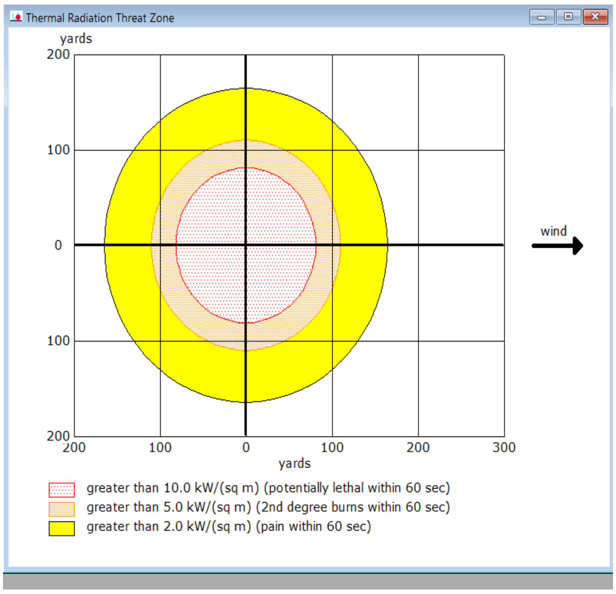


Figure 11 Ethanol _ ALOHA _ Threatzone_Case_1

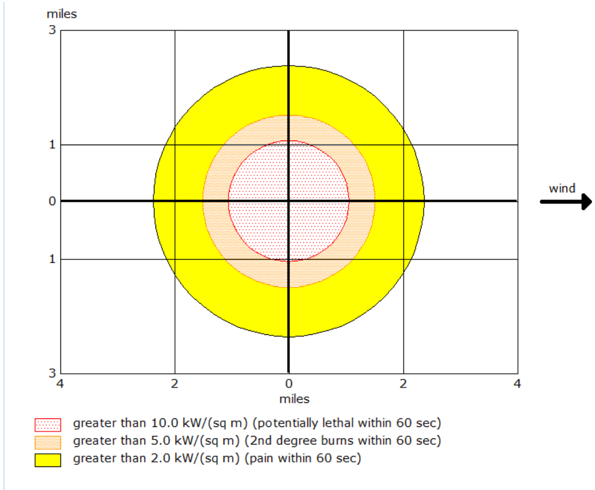


Figure 12 Ethanol_ALOHA_Threatzone_Case_2

5. ***Acetylene:*** [11] [4] [5] [1] [7] [6]
   Thermal Radiation: The Red zone which has an impacted area till 113 yards & Orange zone impact which has an impacted area till 164 yards and yellow zone which has an impacted area till 259 yards.
   Explosion risk: Acetylene has a wide flammability range with 2.5%-80% indicating a significant risk of fire and explosion.
   Health Hazards: Inhalation of acetylene vapours can displace oxygen leading to asphyxiation and is also an asphyxiant itself at very high concentrations.
   Environmental impact: Combustion of acetylene release pollutants such as carbon monoxide and volatile organic compounds can cause air pollution and also potential environmental damage. Spilled acetylene can contaminate soil and water posing risks to aquatic ecosystems.

```
Text Summary
SITE DATA:
   Location: FRANKFURT, GERMANY
   Building Air Exchanges Per Hour: 0.36 (unsheltered single storied)
   Time: June 22, 2024  0345 hours ST (using computer's clock)

CHEMICAL DATA:
   Chemical Name: ACETYLENE
   CAS Number: 74-86-2                        Molecular Weight: 26.04 g/mol
   PAC-1: 65000 ppm    PAC-2: 230000 ppm      PAC-3: 400000 ppm
   LEL: 25000 ppm      UEL: 800000 ppm
   Ambient Boiling Point: -119.4° F
   Vapor Pressure at Ambient Temperature: greater than 1 atm
   Ambient Saturation Concentration: 1,000,000 ppm or 100.0%

ATMOSPHERIC DATA: (MANUAL INPUT OF DATA)
   Wind: 2.2 meters/second from S at 10 meters
   Ground Roughness: open country             Cloud Cover: 5 tenths
   Air Temperature: 22° C                     Stability Class: E
   No Inversion Height                        Relative Humidity: 57%

SOURCE STRENGTH:
   Leak from hole in horizontal cylindrical tank
   Flammable chemical is burning as it escapes from tank
   Tank Diameter: 60 meters                   Tank Length: 75 meters
   Tank Volume: 212,058 cubic meters
   Tank contains liquid                       Internal Temperature: 22° C
   Chemical Mass in Tank: 81,330 tons         Tank is 86% full
   Circular Opening Diameter: 10 centimeters
   Opening is 0.5 meters from tank bottom
   Max Flame Length: 72 yards
   Burn Duration: ALOHA limited the duration to 1 hour
   Max Burn Rate: 38,200 pounds/min
   Total Amount Burned: 2,290,464 pounds
   Note: The chemical escaped from the tank and burned as a jet fire.

THREAT ZONE:
   Threat Modeled: Thermal radiation from jet fire
   Red   : 113 yards --- (10.0 kW/(sq m) = potentially lethal within 60 sec)
   Orange: 164 yards --- (5.0 kW/(sq m) = 2nd degree burns within 60 sec)
   Yellow: 259 yards --- (2.0 kW/(sq m) = pain within 60 sec)
```

Figure 13 Acetylene _ ALOHA _ Data file

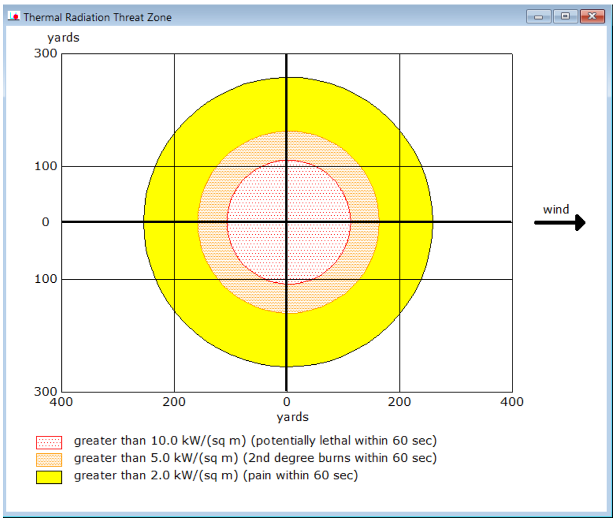


Figure 14 Acetylene _ ALOHA _ Threatzone_Case_1

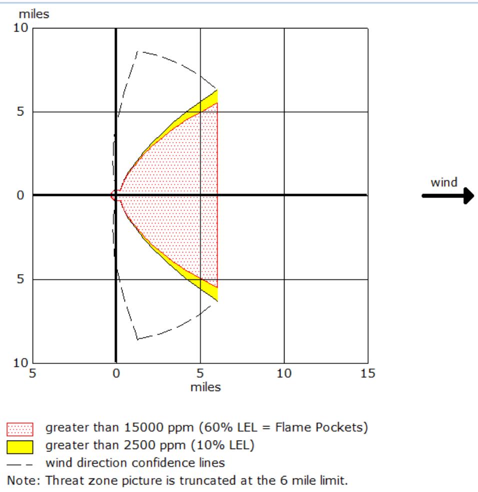
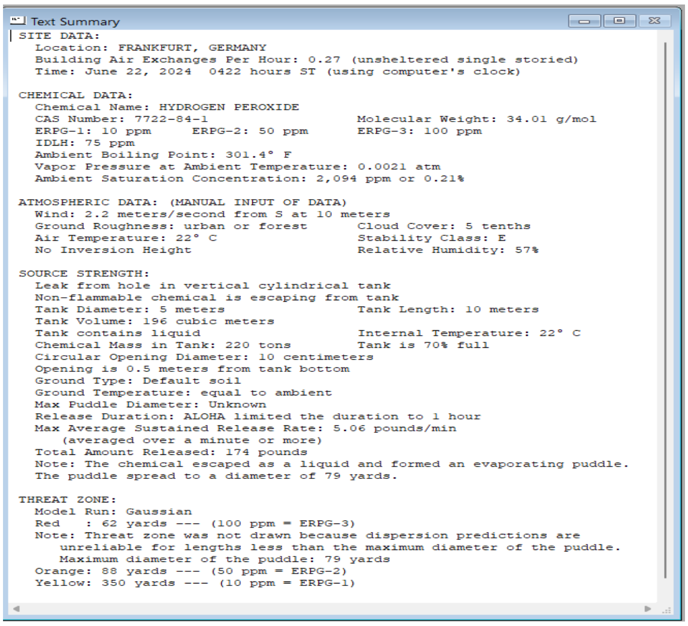


Figure 15 Acetylene_ALOHA_Threatzone_Case_2

6. ***Hydrogen Peroxide:*** [12] [4] [5] [1] [6] [7]
Threat Zone: The Red zone extends up to 62 yards from the source. Hydrogen peroxide concentration at this distance could reach up to 100ppm which corresponds to ERPG-3. Orange zone extends up to 88 yards and concentrations could reach up to 50ppm, corresponding to ERPG-2. Yellow zone extends up to 350 yards and concentrations could reach up to 10ppm, corresponding to ERPG-1.
Environmental Impact: Hydrogen peroxide is not flammable but can decompose into oxygen and water influencing local atmospheric conditions.
Health Hazards: Inhalation of hydrogen peroxide vapours at high concentrations can cause respiratory irritation and other health effects.


```
Text Summary
SITE DATA:
   Location: FRANKFURT, GERMANY
   Building Air Exchanges Per Hour: 0.27 (unsheltered single storied)
   Time: June 22, 2024  0422 hours ST (using computer's clock)

CHEMICAL DATA:
   Chemical Name: HYDROGEN PEROXIDE
   CAS Number: 7722-84-1                     Molecular Weight: 34.01 g/mol
   ERPG-1: 10 ppm      ERPG-2: 50 ppm        ERPG-3: 100 ppm
   IDLH: 75 ppm
   Ambient Boiling Point: 301.4° F
   Vapor Pressure at Ambient Temperature: 0.0021 atm
   Ambient Saturation Concentration: 2,094 ppm or 0.21%

ATMOSPHERIC DATA: (MANUAL INPUT OF DATA)
   Wind: 2.2 meters/second from S at 10 meters
   Ground Roughness: urban or forest         Cloud Cover: 5 tenths
   Air Temperature: 22° C                    Stability Class: E
   No Inversion Height                       Relative Humidity: 57%

SOURCE STRENGTH:
   Leak from hole in vertical cylindrical tank
   Non-flammable chemical is escaping from tank
   Tank Diameter: 5 meters                   Tank Length: 10 meters
   Tank Volume: 196 cubic meters
   Tank contains liquid                      Internal Temperature: 22° C
   Chemical Mass in Tank: 220 tons           Tank is 70% full
   Circular Opening Diameter: 10 centimeters
   Opening is 0.5 meters from tank bottom
   Ground Type: Default soil
   Ground Temperature: equal to ambient
   Max Puddle Diameter: Unknown
   Release Duration: ALOHA limited the duration to 1 hour
   Max Average Sustained Release Rate: 5.06 pounds/min
      (averaged over a minute or more)
   Total Amount Released: 174 pounds
   Note: The chemical escaped as a liquid and formed an evaporating puddle.
   The puddle spread to a diameter of 79 yards.

THREAT ZONE:
   Model Run: Gaussian
   Red    : 62 yards --- (100 ppm = ERPG-3)
   Note: Threat zone was not drawn because dispersion predictions are
      unreliable for lengths less than the maximum diameter of the puddle.
      Maximum diameter of the puddle: 79 yards
   Orange: 88 yards --- (50 ppm = ERPG-2)
   Yellow: 350 yards --- (10 ppm = ERPG-1)
```

Figure 16 Hydrogen Peroxide _ ALOHA _ Datafile

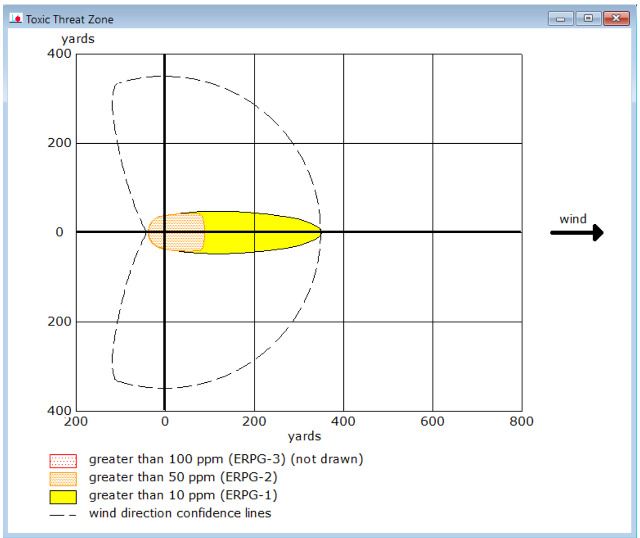


Figure 17 Hydrogen Peroxide _ ALOHA _ Threatzone_Case_1

4.2.2 Domino Effects:

Scenario 1: natural gas leak and explosion

If natural gas leaks and leads to explosion then, due to the heat and shock wave from the explosion could ignite ethanol storage tanks, causing secondary fires or explosion. If the fire reaches acetylene storage, the self-decomposing nature of acetylene could lead to a violent explosion, amplifying the impact. While not directly flammable, the presence of hydrogen peroxide could intensify any fire due to its oxidizing properties.

Scenario 2: Gasoline fire

If gasoline spill leads to a fire, then the fire could spread to biogas storage, causing further explosions and fires due the flammable nature of biogas. The spread of the gasoline fire could ignite ethanol, leading to large flames, which could then heat acetylene tanks. The acetylene could decompose explosively, resulting in a major accident. The fire could be exacerbated by the oxidizing effect of hydrogen peroxide, leading to more intense flames and potential spread to other areas.

Scenario 3: Ethanol explosion

If an ethanol tank ruptures and ignites then the resulting fire could reach the acetylene storage, causing a catastrophic explosion due to acetylene's unstable nature. The explosion could rupture biogas tanks, leading to additional explosions and fires. If the fire spreads to the natural gas areas, it could cause further explosions, potentially affecting a larger part of the plant.

SCENARIO 4: Acetylene explosion

If acetylene decomposes explosively then the explosion would likely ignite nearby ethanol and biogas storage, causing secondary explosions and fires. The shockwave and heat could cause gasoline tank to rupture and ignite, spreading the fire further. The fire's spread could involve hydrogen peroxide, intensifying the flames and leading to more widespread damage.

5. PROTECTION AND INTERVENTION MEASURES

From the risk analysis data generated by ALOHA it is clear that comprehensive safety measures are required for storage of this hazardous substances and this measures which reduce the impact of risks and ensure safety.

5.1 Substance specific safety measures

Natural gas:

- Ensure proper ventilation to storage tanks to avoid gas accumulation.
- Ensure the storage tank is equipped with pressure relief valves to prevent over pressurisation.
- Ensure the Storage of Natural gas is not near the flammable liquids.
- Installation of leak detectors & alarms helps in identifying the undesired event.

Biogas:

- Storage of biogas in sealed tanks to maintain anaerobic condition and which in turn prevents accidental ignition.
- Monitoring pressure and temperature is very crucial
- Installation of flares which helps in safely burn off the excess biogas.

Ethanol:

- Installation of Bund walls around storage tanks acts as a spill containment.
- Ensuring adequate ventilation is provided to disperse vapours and prevent explosive atmosphere.
- Foam based fire suppression systems are sutabile for Ethanol storage tanks.

Gasoline:

- Ensure proper grounding and bonding to storage tanks to avoid static discharge.
- Providing explosion proof electrical equipment for gasoline storage tanks.
- To avoid gasoline vapor losses, implement vapor recovery systems.

Acetylene:

- Ensure acetylene storage tank is separated from oxidisers and flammable substances.
- Continuously monitor the temperature conditions of acetylene storage tank.
- Avoid storage of acetylene near or at high temperatures.
- Provide necessary safety layers with respect to design to avoid decomposition.

Hydrogen peroxide:

- Storage of H202 should be very crucial to keep away from sunlight, cool and well-ventilated areas.
- Ensure the material compatibility to avoid decomposition.

5.2 General Safety Measures

1. **Emergency response Planning:** Maintenance of Emergency response plans which includes evacuation, firefighting, spill containment and communication procedures with emergency services. Regular drills should be conducted with the facility personnel.
2. **Safety Equipment:** Availability and accessibility of PPE (Personal protection equipment) should be maintained. Installation of fire detection and suppression systems be available.
3. **Access Control:** Strict access control measures to restrict unauthorised personnel should be taken from entering storage areas. Storage area should be under full time surveillance.
4. **Training:** Providing regular training to the personnel on handling, storage and emergency procedures**.**
5. **Signage:** Mark all the storage areas with appropriate hazard signs and safety instructions and also emergency exits should be clearly marked.

By implementing this safety measures, the risk associated with the storage of this hazardous substances “Natural gas, Biogas, Ethanol, Gasoline, Acetylene and Hydrogen peroxide can be effectively managed, and ensuring the safety of the people, facility and the environment. [2]

6. CONCLUSION

According to the safety study, the chemical production facility is classified as an upper-tier establishment under the [2]Seveso-III directive because it handles several dangerous substances which exceed the upper-tier thresholds. The facility is exposed to great risks such as fire, explosions and release of toxic materials calling for extensive safety precautions. These may involve specific safety protocols for each substance, general safety measures as well as strong emergency response plans. In order to reduce these hazards, the plant needs to enforce strict safety controls, install continuous monitoring systems and develop effective emergency preparedness stratergies.it is important also to train the staff regularly on safety measures while liaising with local fire brigade among other emergency services. By adhering to the SEVESO-III directive's requirements, the facility can significantly reduce the likelihood and impact of major accidents, safeguarding human health and the environment.

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